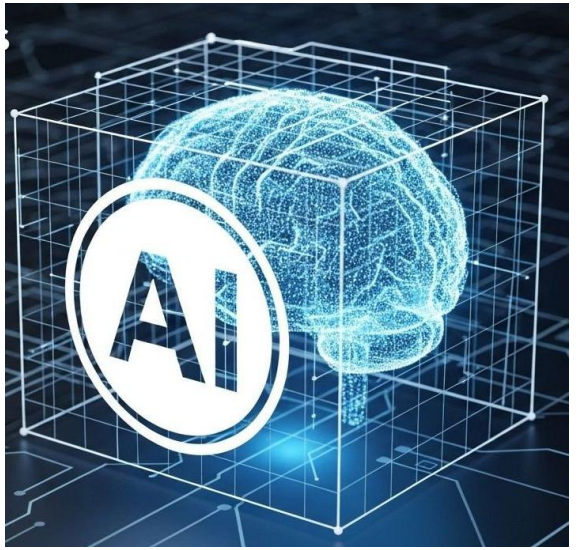



# ALY 3510 · Fall 2026

# Lecture 9: AI & Creativity (additional slides)

*Can machines create — and what does that mean for us?*

# Additional Materials for Self-Learning

# Computational Creativity

### Construct
Build programs capable of human-level creativity

### Understand
Study human creativity through algorithms

### Enhance
Design tools that amplify human creativity

## The AI Creative Cycle

1 → 2 → 3 → 4

**Ideation**
Analyze patterns

**Generation**
Produce outputs

**Refinement**
Iterate & improve

**Realization**
Final creation

# Where AI Helps Most

**~25%** productivity increase for artists using text-to-image tools

## Research Insights

- Peak novelty increases as artists master AI tools
- Average novelty narrows—risk of homogenization
- Human-first ideation preserves creative diversity

*Source: Zhou & Lee (2024)*




**Best Practice**

Use AI as a

## CO-PILOT

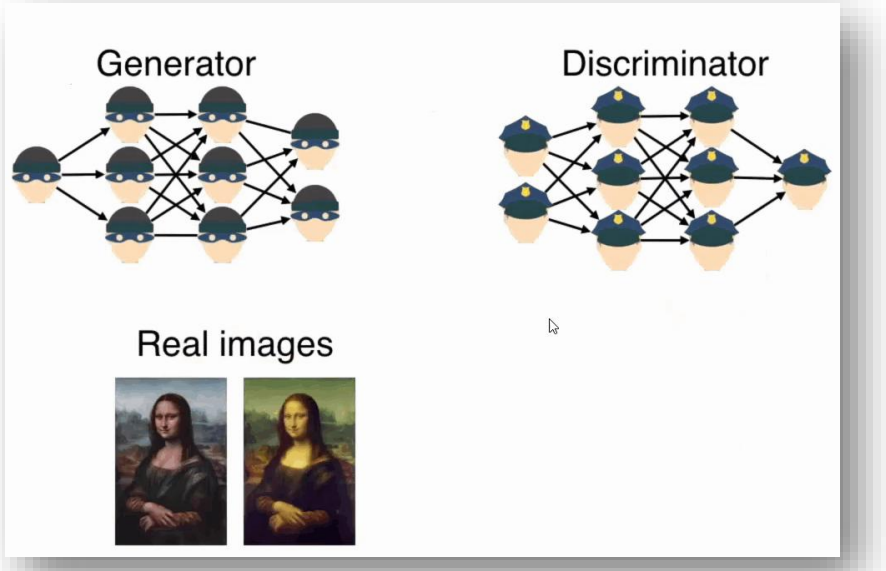
Humans ideate first,
AI refines and generates variations

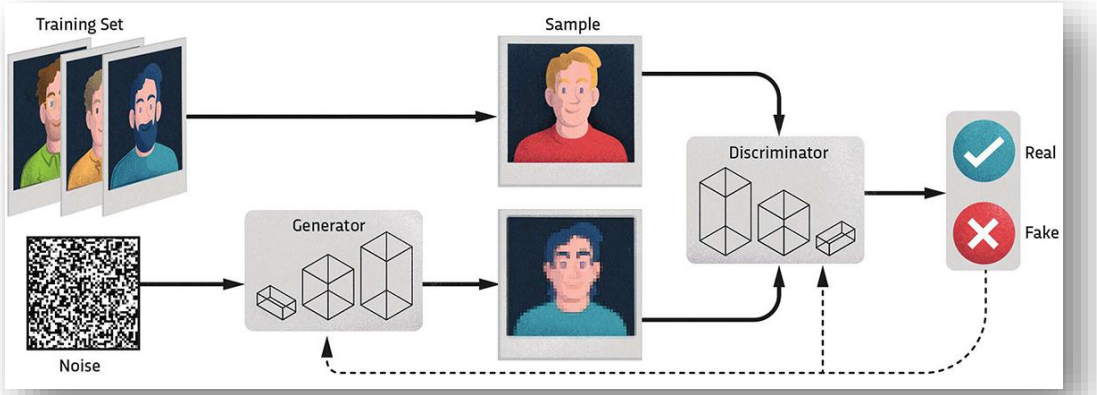
***AI amplifies—not replaces***

# How AI creates: Key Technologies (GANs)

## Generative Adversarial Networks (GANs)

- Two neural networks compete - one creates content, the other evaluates it. Through this "adversarial" process, the creator improves until the outputs are realistic.




https://share.google/images/o427g6fUFuQP2KkEM
https://share.google/images/6MgGqPMFR7ZZeDd2T

# How Stable Diffusion Works

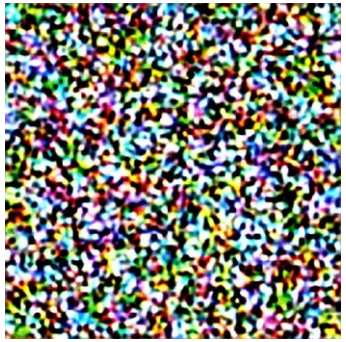
## Forward Process (Training)

1. Start with a clear image
2. Add noise incrementally
3. Train the model to predict noise
4. Result: A noise predictor

Image → Noisy → Noisier → Pure Noise

## Reverse Process (Generation)

1. Start with random noise
2. Predict & subtract noise
3. Guide with text prompt
4. New image emerges!



Noise → Clearer → Clearer → Image!

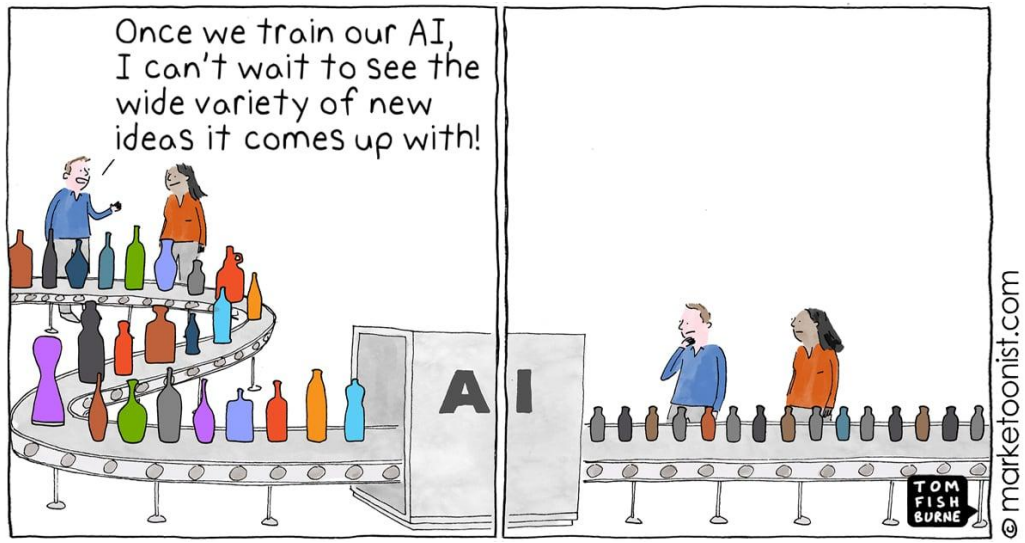
## The Key Insight

By learning to predict noise, the model learns the structure of images. During generation, it "imagines" what image could have produced the noise pattern, then subtracts noise to reveal that image.

# The Novelty Paradox

**Peak novelty ↑ but Average novelty ↓**

The best AI art is more creative than ever, but most AI art looks the same.




## Implications

- Creative space is expanding, but with inefficiencies
- Risk of homogenization in AI-generated content
- Success requires strong ideation and filtering skills
- Human creativity and judgment remain essential

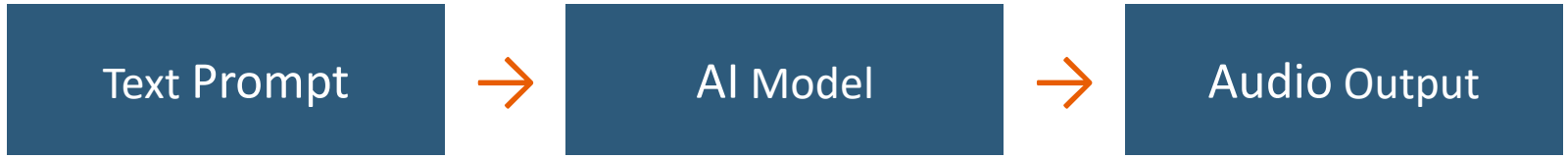
Zhou & Lee (2024); Aher et al. (2024), "Generative AI enhances individual creativity but reduces collective diversity"

# AI Music Generation: Composing the Future

| Suno AI | AIVA | Udio | MusicFX |
| --- | --- | --- | --- |
| Complete songs with lyrics and vocals from text prompts | Emotional soundtracks for films and games | "ChatGPT for music" - custom tracks from descriptions | Google's most accurate text-to-song generation |

**Democratizing music creation: non-musicians can generate professional-quality compositions**

**How It Works**



# AI Tools for Non-Technical Creators

## Accessibility Features

| | |
|---|---|
| **No Coding Required** | Natural language interfaces |
| **Template Systems** | Pre-built starting points |
| **Real-time Feedback** | Instant generation |
| **Cross-Platform** | Works with existing software |

## Democratization of Creativity

AI tools make professional-quality creative output accessible to everyone

**Before AI**

Years of training

→

**With AI**

Minutes to create

# Ethical Concerns: Bias in AI Creativity

## Identified Problems

| | |
|---|---|
| **Dataset Bias** | Training reflects historical prejudices |
| **Underrepresentation** | Groups marginalized in outputs |
| **Cultural Homogentic** | Tendency toward mainstream aesthetics |
| **Algorithmic Amplify** | Biases strengthened in generation |

## Real-World Example

Image searches for different ethnicities often return stereotypical content, revealing deep biases in training data.

*The "librarian" stereotype: AI generates predominantly older white women for librarian images.*

# The Future of AI Creativity Tools

## Emerging Trends

### Multimodal Generation

Creating synchronized audio, visual, and text

### Real-time Collaboration

Live creative sessions between humans and AI

### Personalized AI Artists

Systems that learn individual creative styles

### Ethical AI Models

Tools built with consent and attribution

**Prediction**

Within a decade, AI creative tools will be as common as word processors.

# Best Practices for Ethical AI Creativity

## Guidelines for Responsible Use

| | |
|---|---|
| ✓ **Transparency** | Always disclose AI use |
| ✓ **Attribution** | Credit human and AI contributions |
| ✓ **Consent** | Use tools respecting artists' rights |
| ✓ **Augmentation** | Enhance, don't replace creativity |
| ✓ **Critical Evaluation** | Question biases and limitations |

## Professional Standards

Industries developing AI guidelines:

- Film & Television
- Music Production
- Publishing
- Graphic Design
- Gaming

**Goal:** Use AI responsibly to amplify human creativity while respecting creators' rights.

# Building Creative AI Literacy

## Essential Skills for the AI Creative Era

| | |
|---|---|
| **Prompt Engineering** | Crafting effective instructions for AI systems |
| **Critical Evaluation** | Assessing the quality and appropriateness of AI output |
| **Ethical Awareness** | Understanding implications of AI use |
| **Hybrid Workflows** | Integrating AI tools with traditional methods |
| **Creative Direction** | Guiding AI toward meaningful outcomes |

## Future Career Opportunity

"AI Creative Direction" may become a new professional field, requiring both artistic vision and technical understanding.

# Scientific Creativity: Definition & Applications

**Scientific creativity: The ability to generate novel ideas, approaches, or solutions within science, leading to new discoveries, theories, or technologies.**

## Future Scientific Applications

- **Hypothesis Generation** — AI proposes novel scientific theories
- **Experimental Design** — Optimizing experiments for maximum learning
- **Cross-disciplinary Insights** — Connecting ideas across scientific fields
- **Autonomous Labs** — AI-controlled experiments running 24/7
- **Autonomous Chemistry** — AI designs and executes experiments independently

# AlphaFold: A Scientific Revolution

### The 50-Year Problem

Predicting how proteins fold had stumped scientists since the 1970s.

*Traditional: months to years per protein*

### 🏆 2024 Nobel Prize

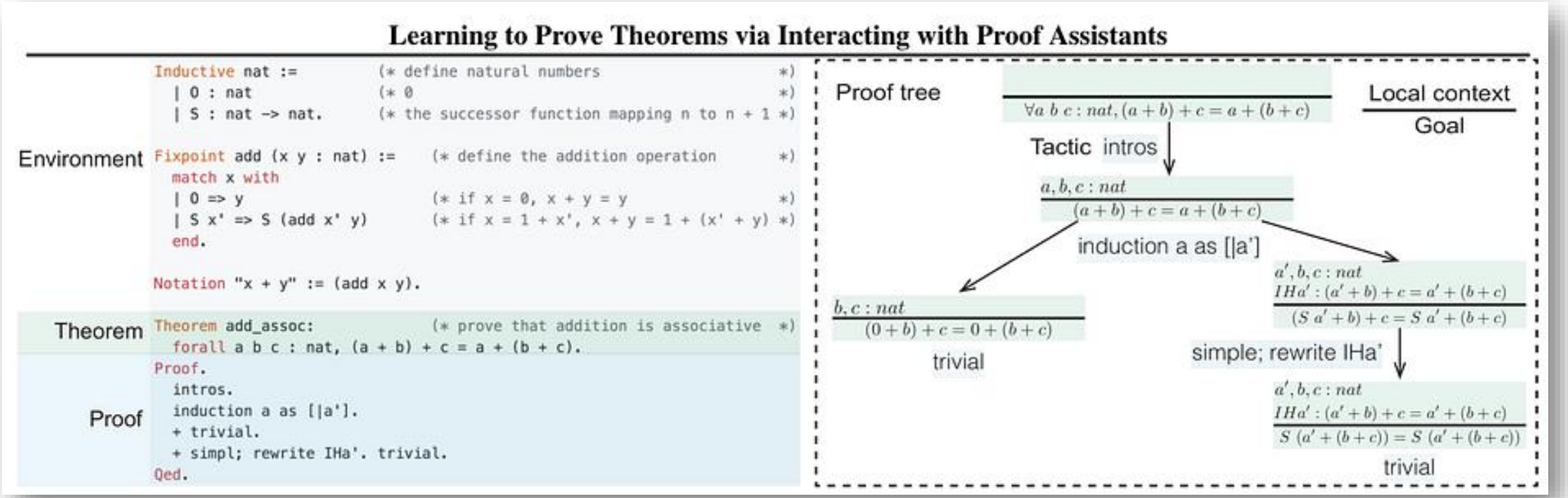
Google DeepMind's Demis Hassabis & John Jumper awarded Nobel Prize in Chemistry for AlphaFold

## Impact

| 200M+ | 2M+ | 190 | Minutes |
|---|---|---|---|
| Proteins predicted | Researchers using it | Countries accessing | Instead of months |

# Automated Theorem Provers (ATPs)

AI systems that can generate mathematical proofs, check proof validity, and even solve competition-level mathematics problems.




https://tinyurl.com/285dvamu

# AI Achieves Gold at Math Olympiad

## 🥇 Historic Achievement

DeepMind and OpenAI systems claim gold-medal performance at the International Mathematical Olympiad in 2025

*Competing against the world's brightest young mathematicians*

### The Challenge

Unlike chess, with a finite number of moves, theorem proving has an infinite number of possible approaches. AI must learn creative problem-solving strategies similar to human mathematicians — this represents a fundamentally different kind of intelligence.